

Design and Implementation of Smart Malaria Diagnosis Assistant in Rural Clinics.



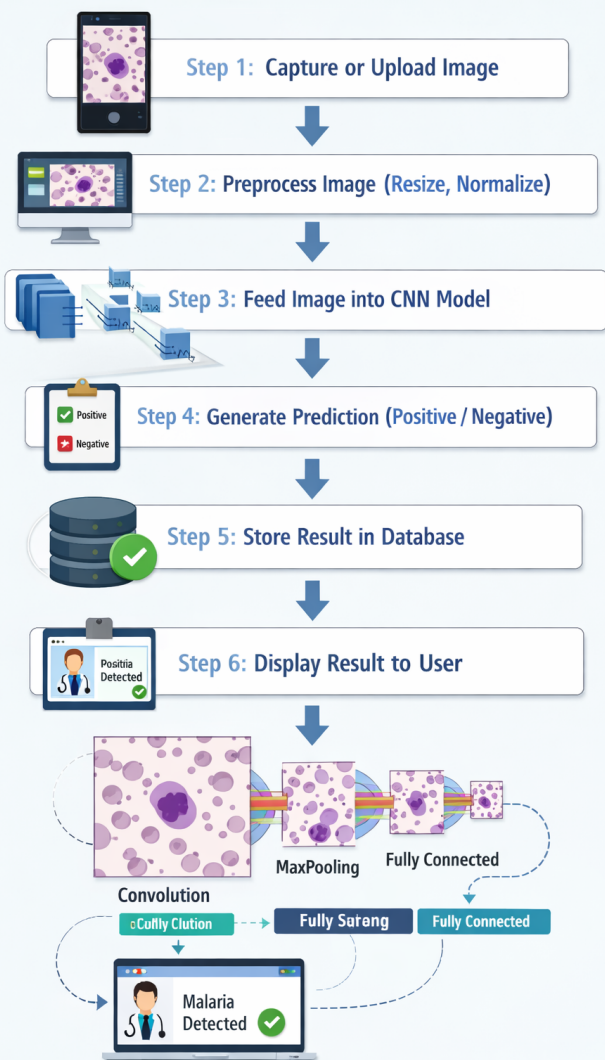
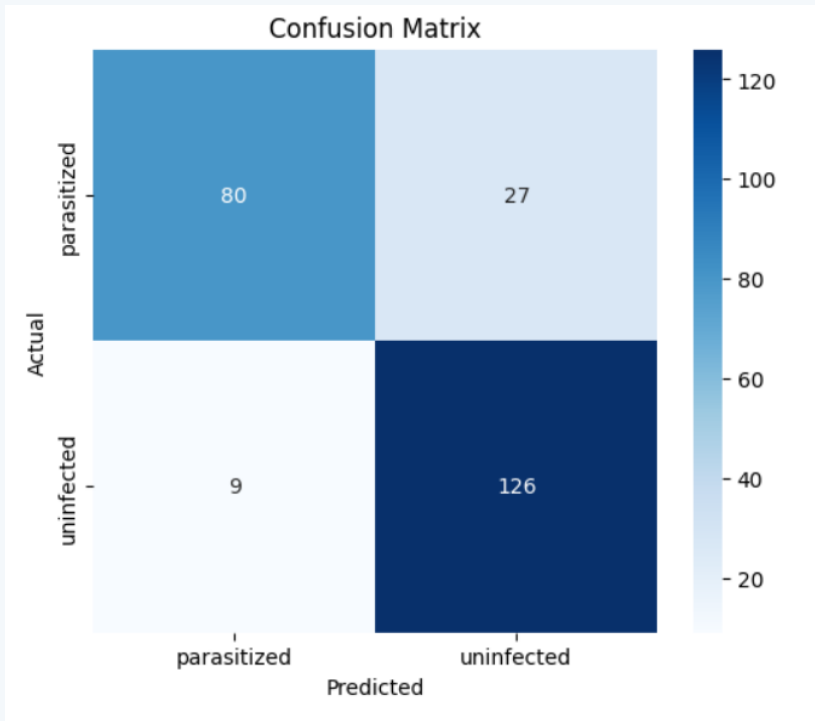
Student: Agar Marial Riak | **Supervisor:** Eng. Thon Kuir

Problem & Motivation

- **Problem Context:** Manual microscopy is slow, specialist-dependent, and difficult to scale in resource-constrained facilities.
- **Research Gap:** Rural laboratories need a lightweight decision-support tool integrated with patient records and clinical review.
- **Objective:** Classify thin-smear cell images as parasitized or uninfected and return an interpretable confidence score.

Evaluation & Key Results

85% test accuracy



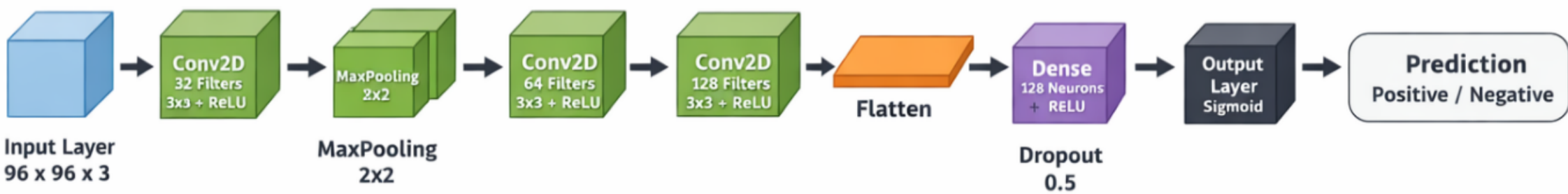
Methodology & Dataset

System Architecture

Our system utilizes **CNN**, depicted in the diagram below, utilizing transferring from a **MobileNetV2** backbone pre-trained on **ImageNet**. This system extract features from **RGB** images and performs **binary classification** to determine wether malaria is detected or not.

Dataset

Our data is sourced from **NIH/NLM malaria dataset** consisting of **27,558** balanced parasitized and uninfected cell images.



Significance & Conclusion

- **Contribution:** An end-to-end web assistant combining CNN inference, patient records, role-based review, and PDF reporting.
- **Impact:** Supports laboratory screening and consistent documentation; it assists—not replaces—qualified clinicians.
- **Future Work:** Patient-level external validation in South Sudan, explainability, and offline mobile deployment.